

AI-Based Research Paper Recommendation System

Query : Brain Tumor Detection using Deep Learning

Recommended Research Paper

Implementation of deep learning algorithm for automatic detection of brain tumors using intraoperative IR-thermal mapping data

Similarity Score : 0.6452

Abstract

The efficiency of deep machine learning for automatic delineation of tumor areas has been demonstrated for intraoperative neuronavigation using active IR-mapping with the use of the cold test. The proposed approach employs a matrix IR-imager to remotely register the space-time distribution of surface temperature pattern, which is determined by the dynamics of local cerebral blood flow. The advantages of this technique are non-invasiveness, zero risks for the health of patients and medical staff, low implementation and operational costs, ease and speed of use. Traditional IR-diagnostic technique has a crucial limitation - it involves a diagnostician who determines the boundaries of tumor areas, which gives rise to considerable uncertainty, which can lead to diagnosis errors that are difficult to control. The current study demonstrates that implementing deep learning algorithms allows to eliminate the explained drawback.

Generated Summary

The efficiency of deep machine learning has been demonstrated for intraoperative neuronavigation using activeIR-mapping with the use of the cold test. The advantages of this technique are non-invasiveness, zero risks for health of patients and medical staff, low implementation and operational costs, ease and speed of use.

Detected Named Entities

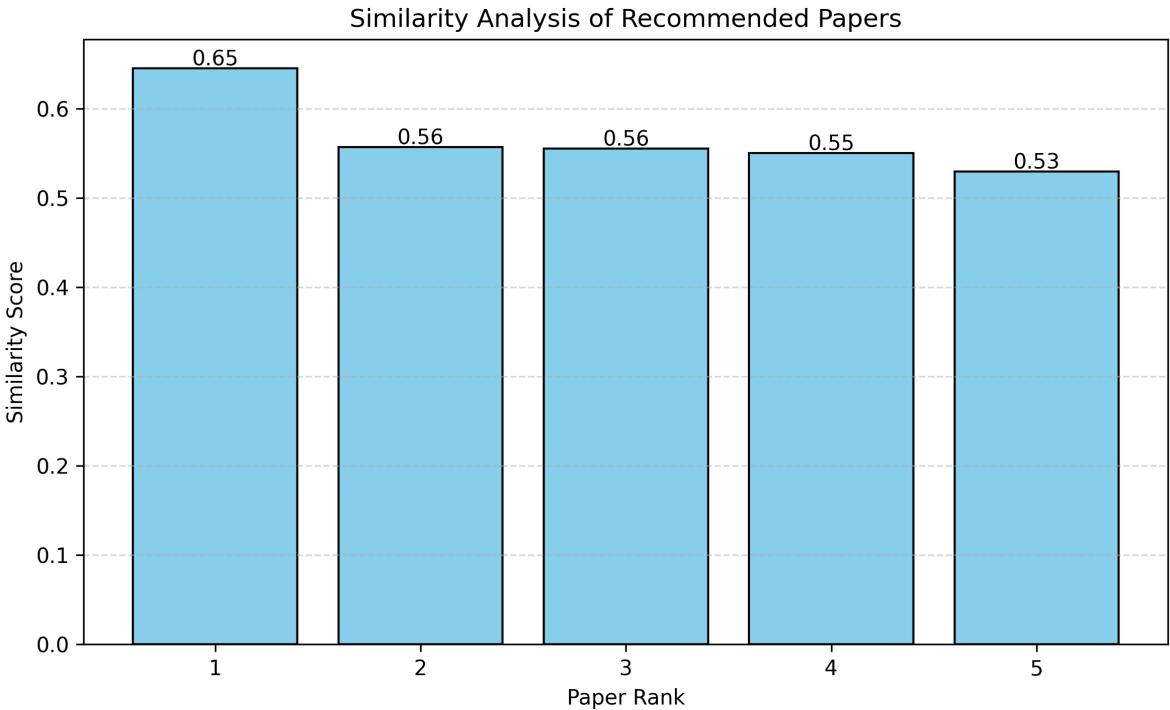
Entity	Label
activeIR	ORG
zero	CARDINAL

Important Keywords

Keyword	Score
intraoperative neuronavigation using	0.650

intraoperative neuronavigation	0.599
neuronavigation using activeir	0.592
demonstrated intraoperative neuronavigation	0.582
learning demonstrated intraoperative	0.474

Similarity Score Analysis



Keyword Distribution

Most Relevant Keywords

active learning
neuronavigation
demonstrated
intraoperative
using

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